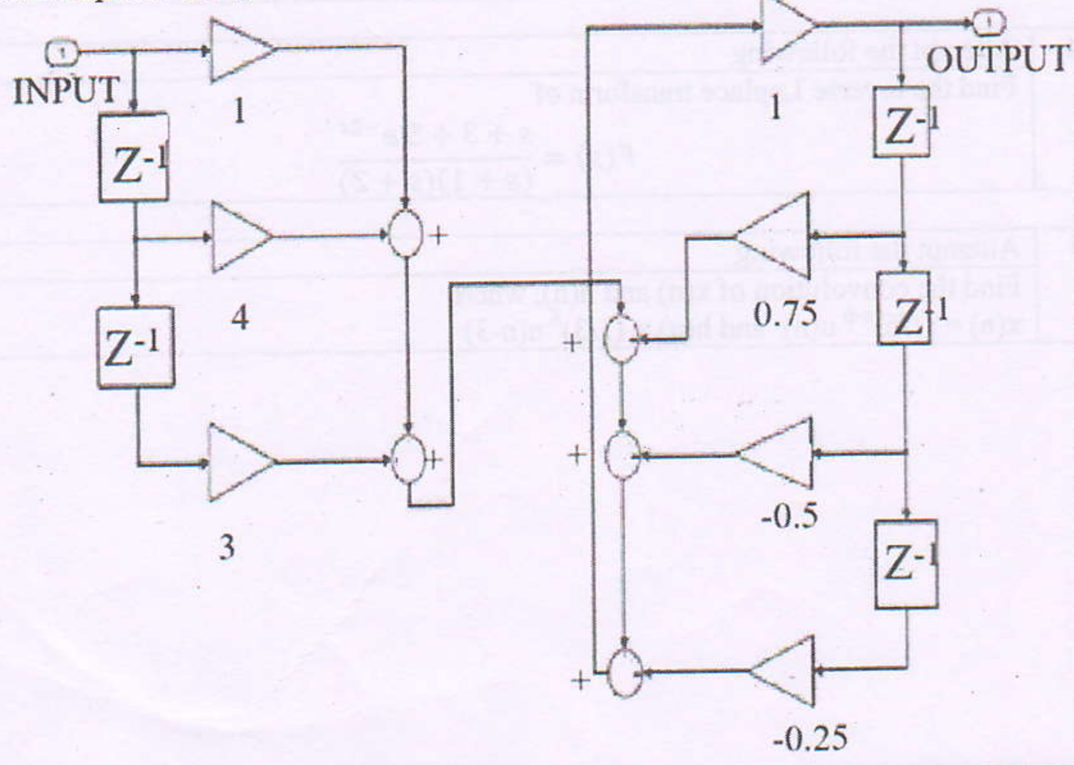
SOMAIYA VIDYAVIHAR UNIVERSITY

08.05.2025(E)

Semester: January 2025– April 2025			
Maximum Marks: 50	**Examination: End-Semester Examination**		**Duration: 2 Hrs.**
Programme code: 41 **Programme: B. Tech in Computer and Communication Engineering**	**Class: SY**		**Semester: IV** (SVU-2023)
Institute/School/Department: K. J. Somaiya School of Engineering	**Name of the department/Section/Centre:** EXTC		
Course Code: 216U41C404	**Name of the Course: Signals and Systems**		
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **two**	**10**
i)	a) Simplify the expression $[e^{-t}\cos(3t-60^o)]\delta(t)$ b) Evaluate $\int_{-\infty}^{+\infty}\delta(\tau)f(t-\tau)d\tau$ c) Show that $\int_{-\infty}^{+\infty}\delta(t)\,e^{-j\omega t}\,dt = 1$ d) Given f(t)= 1, $1\le t\le 3$, 0, otherwise. Find f (2t – 6). e) Give mathematical expression of any one even signal.	**05**
ii)	Find ouput of the system in the form of difference equation.  	**05**

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iii)	Use the Z transform to perform the convolution of the following two sequences. $H(n) = (1/2)^n \quad 0 \le n \le 2$ $0, \quad$ otherwise $X(n) = \delta(n) + \delta(n-1) + 4\,\delta(n-2)$	**05**
Q.2	Attempt any **one**	**10**
i)	If the input x(n) to LTI Discrete system is, $x(n) = (1/2)^n u(n) + 2^n u(-n-1)$ the output is $y(n) = 6\,(1/2)^n u(n) - 6\,(3/4)^n u(n)$. Find system function H(z).	
ii)	A causal LTI system is characterized by the difference equation $y(n) = 0.25\, y(n-1) + 0.125\, y(n-2) + x(n) - x(n-1)$. Find the system function H(z) and unit impulse response h(n).	
Q.3	Attempt any **one**	**10**
i)	Find the Fourier series of the signal f(t), In one period –T/2 to T/2, the signal f(t) is expressed $(\tau/2 < T/2)$ $f(t) = 1, \; -\tau/2 \le t \le \tau/2$ 0, otherwise.	
ii)	Obtain the Fourier transform of the rectangular pulse $g_T(t) = 1, \; \lvert t \rvert \le T/2$ 0, otherwise.	
Q.4	Attempt the following	**10**
	Find the inverse Laplace transform of $$F(s) = \frac{s + 3 + 5\,e^{-2s}}{(s+1)(s+2)}$$	
Q.5	Attempt the following	**10**
	Find the convolution of x(n) and h(n), where $x(n) = (1/6)^{n-6} u(n)$ and $h(n) = (1/3)^n u(n-3)$	